

On the existence of vertex transitive 4-ordered hamiltonian and 4-ordered hamiltonian connected cubic graphs

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Abstract¹

A graph is *k*-ordered hamiltonian if for every ordered set S of k vertices there exists a hamiltonian cycle which encounters S in its designated order. A graph is *k*-ordered hamiltonian connected if for every k ordered set $S = \{x_1, x_2, \dots, x_k\}$ there exists a hamiltonian path joining x_1 to x_k that encounters S in its designated order. In this paper, we prove that the five dimensional cube-connected cycles graph, CCC_5 , is 4-ordered hamiltonian and 4-ordered hamiltonian connected graphs. This is the first non-trivial vertex transitive graph with this property.

Index Terms—*k*-ordered hamiltonian; *k*-ordered hamiltonian connected graphs; cube connected cycles graph

I. INTRODUCTION

For the graph definitions and notation, we follow the definitions and notation of [7]. Let $G = (V, E)$ be a graph if V is a finite set and E is a subset of $\{(u, v) \mid (u, v) \text{ is an unordered pair of } V\}$. We say that V is the *vertex set* and E is the *edge set*. Two vertices u and v are *adjacent* if $(u, v) \in E$. The *degree* of a vertex u in G , denoted by $\deg_G(u)$, is the number of vertices adjacent to u . A graph G is *k*-regular if $\deg_G(x) = k$ for any $x \in V$. A *path* is a sequence of adjacent vertices, written as $\langle v_0, v_1, \dots, v_m \rangle$, in which all the vertices

$v_0, v_1, \dots, v_m$ are distinct except for the possibility that $v_0 = v_m$. A *vertex-transitive graph* is a graph such that every pair of vertices is equivalent under some element of its automorphism group. Similarly, An *edge-transitive graph* is a graph such that any two edges are equivalent under some element of its automorphism group.

A *cycle* is a path with at least three vertices such that the first vertex is the same as the last one. A *hamiltonian cycle* is a cycle spans G . A *hamiltonian path* is a spanning path that joins two different vertices of G . A graph G is *hamiltonian-connected* if for every pair u, v of distinct vertices, G contains a $u - v$ hamiltonian path. Every hamiltonian-connected graph of order at least 3 is hamiltonian.

Recently, a new strong hamiltonian property was introduced by Ng and Schultz in [14]. A graph is *k*-ordered hamiltonian if for every k ordered set $S = \{x_1, x_2, \dots, x_k\}$ there exists a hamiltonian cycle which encounters S in its designated order. Obviously, any hamiltonian graph is 3-ordered hamiltonian. Thus, the *k*-ordered hamiltonian property is really interesting only for $k \geq 4$.

A graph is *k*-ordered hamiltonian connected if for every k ordered set $S = \{x_1, x_2, \dots, x_k\}$ there exists a hamiltonian path joining x_1 to x_k that encounters S in its designated order. Note that *k*-ordered

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hamiltonian connected does not imply k -ordered hamiltonian. Moreover, any hamiltonian connected graph is 3-ordered hamiltonian connected. Thus, k -ordered hamiltonian connected is really interesting only for $k \geq 4$.

In this paper, we prove that the five dimensional cube connected cycles graph, CCC_5 , is 3-regular, vertex transitive, 4-ordered hamiltonian, and 4-ordered hamiltonian connected. In the following section, we review the background. We give our main result in section 3. Finally, we give our concluding remarks.

II. REVIEW

Note that the k -ordered hamiltonian property for a graph G will force G to have sufficient connectivity. For example, assume that G has a $(k-2)$ -cutset $K = \{x_1, x_2, \dots, x_{k-2}\}$. Let x_{k-1} and x_k be any two vertices from different components of $G - K$. We set S as $\{x_1, x_2, \dots, x_k\}$. Obviously, there is no cycle that contains S . Thus, any k -ordered hamiltonian graph is $(k-1)$ -connected if $k \geq 3$.

Similarly, assume that G has a $(k-2)$ -cutset $K = \{x_1, x_2, \dots, x_{k-2}\}$. Let x_{k-1} and x_k be any two vertices from different components of $G - K$. We set S as $\{x_1, x_2, \dots, x_k\}$. Obviously it will be impossible for a path P to contain S joining x_1 to x_k . Again, any k -ordered hamiltonian connected graph is $(k-1)$ -connected if $k \geq 3$.

In [14], Ng and Schultz proposed four interesting directions: (1) the sufficient condition for graphs being k -ordered hamiltonian, (2) the existence of small degree k -ordered hamiltonian graphs, (3) the sufficient condition for graphs being k -ordered hamiltonian connected, and (4) the existence of small degree k -ordered hamiltonian connected graphs. There are a lot of studies on the directions (1) and (3) [2–6, 10–12]. However, there have been no studies on the directions (2) and (4) for more than ten years.

It is obvious that we should find 4-ordered hamiltonian and 4-ordered hamiltonian connected graphs

among 3-regular graphs. For a while, K_4 and $K_{3,3}$ were the only two examples of 3-regular 4-ordered hamiltonian graphs. Obviously, K_4 is 4-ordered hamiltonian connected whereas $K_{3,3}$ is not.

Instead of considering 3-regular 4-ordered hamiltonian graphs, Mészáros [13] studied 3-regular 4-ordered graphs in 2008. A graph G on n vertices, $n > 3$, is k -ordered for any integer k , $2 \leq k \leq n$, if for every k -ordered set $S = \{x_1, x_2, \dots, x_k\}$ in G , there exists a cycle, not necessary hamiltonian, that contains all the vertices of S in the designated order. Mészáros [13] proved that the Petersen graph and the Heawood graph are 4-ordered 3-regular graphs. Moreover, the generalized Honeycomb torus $GHT(3, n, 1)$ is 4-ordered if n is an even integer with $n \geq 8$. Furthermore, the Heawood graph is 4-ordered hamiltonian. Using a construction scheme, Tsai et. al. [17] proved that there exists a bipartite non-vertex-transitive 4-ordered 3-regular graph of order n for any sufficiently large even integer n . Moreover, there exists a non-bipartite non-vertex-transitive 4-ordered 3-regular graph of order n for any sufficiently large even integer n . Thus, we have a lot of examples of 3-regular 4-ordered graphs.

Recently, Hsu et. al. [8] classified those generalized Petersen graphs $P(n, 3)$ to be 4-ordered hamiltonian. Assume that n and k are positive integers with $n \geq 2k + 1$. The *generalized Petersen graph* $P(n, k)$ is the graph with vertex set $\{i \mid 0 \leq i \leq n-1\} \cup \{i' \mid 0 \leq i \leq n-1\}$ and edge set $\{(i, i+1) \mid 0 \leq i \leq n-1\} \cup \{(i, i') \mid 0 \leq i \leq n-1\} \cup \{(i', (i+k)') \mid 0 \leq i \leq n-1\}$ where the indices are always taken modulo n . It is proved that the generalized Petersen graph $P(n, 3)$ is 4-ordered hamiltonian if and only if n is even and either $n = 18$ or $n \geq 24$. Thus, there exists an infinite number of 3-regular 4-ordered hamiltonian graphs.

Note that $P(n, 3)$ is vertex transitive if and only if $n = 8$ or $n = 10$. Can we find an infinite number of 3-regular vertex transitive 4-ordered hamiltonian

graphs? Sherman et. al. [16] answer the question affirmatively by considering a family of graphs which is a generalization of the Heawood graph. The *Chordal Ring graph*, denoted by $CR(2n, 1, k)$ with $3 \leq k \leq n$, is a graph with its vertex set $\{v_i \mid 0 \leq i < 2n\}$ and edge set $\{(v_i, v_{i+1}) \mid 0 \leq i < 2n\} \cup \{(v_i, v_{i+k}) \mid i \text{ is even and } 0 \leq i < 2n\}$ where the indices are always taken modulo $2n$. Note that the Heawood graph is $CR(14, 1, 5)$. Moreover, any Chordal ring graph is vertex transitive. It is proved that $CR(2n, 1, 5)$ is 4-ordered hamiltonian if and only if $2n = 12k + 2$ or $2n = 12k + 10$ for some $k \geq 2$ or $2n = 14$.

Thus, there is some progress in direction (2). What is the progress in direction (4)? We note that $P(n, 3)$ is a bipartite graph if n is even. Moreover, $CR(2n, 1, k)$ is a bipartite graph if $3 \leq k \leq n$. However, any bipartite graph except K_1 and K_2 is not hamiltonian connected. The counterpart terminology of hamiltonian connected in bipartite graph is known as hamiltonian laceable. For this reason, we have the concept of k -ordered hamiltonian laceable. A bipartite graph G is *k -ordered hamiltonian laceable* if for any sequence of k distinct vertices $t_1, t_2, \dots, t_k$ of G where t_1 and t_k are in different partite sets, there exists a hamiltonian path between t_1 and t_k in G containing these k vertices in the specified order. It is interesting to point out in [8] that $P(n, 3)$ is 4-ordered hamiltonian laceable if and only if $n \geq 10$ and n is even.

We may apply the same technique in [16] to obtain the following theorem.

Theorem 2.1: $CR(2n, 1, 5)$ is 4-ordered hamiltonian and 4-ordered hamiltonian laceable if and only if $2n = 12k + 2$ or $2n = 12k + 10$ for some $k \geq 2$.

Thus, the Heawood graph, $CR(14, 1, 5)$, is the only graph in the family $\{CR(2n, 1, 5) \mid n \geq 5\}$ that is 4-ordered hamiltonian but not 4-ordered hamiltonian laceable. Obviously, we have an infinite number of vertex transitive 3-regular graphs that are 4-ordered hamiltonian and 4-ordered hamiltonian laceable. However, using a computer, we can check

that $CR(34, 1, 5)$, $CR(40, 1, 5)$, and $CR(46, 1, 5)$ are 4-ordered hamiltonian laceable. Note that these three graphs are not 4-ordered hamiltonian.

Now, can we find a 3-regular 4-ordered hamiltonian connected graphs except K_4 ? In [8], it is proved that $P(n, 3)$ is 4-ordered hamiltonian connected if and only if n is odd and either $n = 15$ or $n \geq 19$. We note that graphs in this family is 4-ordered hamiltonian connected but not 4-ordered hamiltonian.

Can we find an infinite family of 3-regular graphs that are both 4-ordered hamiltonian and 4-ordered hamiltonian connected? Again, the answer is affirmative. Hung et al. [9] prove that $P(n, 4)$ is 4-ordered hamiltonian if and only if $n \geq 18$ and $n \neq 20$ and $P(n, 4)$ is 4-ordered hamiltonian connected if and only if $n \geq 18$. We note that $P(20, 4)$ is 4-ordered hamiltonian connected but not 4-ordered hamiltonian.

We note that $P(n, 4)$ is vertex transitive if and only if n is 15 or 17. Thus, K_4 is the only known vertex transitive graph that is both 4-ordered hamiltonian and 4-ordered hamiltonian connected. In this paper, we present a non-trivial vertex transitive graph that is both 4-ordered hamiltonian and 4-ordered hamiltonian connected.

III. MAIN RESULT

The cube-connected cycles graph CCC_n was introduced by Preparata and Vuillemin [15] as a good alternate for the hypercube, having a fixed degree equal to 3 and yet a small diameter compared to its number of vertices. This graph has since been studied by many people, who showed in particular, that it also has good properties as far as communications are concerned.

Let $\mathbf{u} = u_{n-1}u_{n-2} \dots u_1u_0$ be an n -bit binary string. We use $(\mathbf{u})_j$ to denote the bit u_j . We use $(\mathbf{u})^j$ to denote the vertex $\mathbf{v} = v_{n-1}v_{n-2} \dots v_1v_0$ with $(\mathbf{v})_k = (\mathbf{u})_k$ if and only if $k = j$. The graph CCC_n has $n \times 2^n$ vertices, labeled $(l, \mathbf{x})$, where l is an integer between 0 and $n-1$ and $\mathbf{x}$ is an n -bit binary

string. Two vertices $(l, \mathbf{x})$ and $(l', \mathbf{y})$ are adjacent if and only if either $\mathbf{x} = \mathbf{y}$ and $|l - l'| = 1$, or $l = l'$ and $\mathbf{y} = (\mathbf{x})^l$.

Using a computer, we have the following result. The program and the output files are available at <http://dl.dropbox.com/u/3554341/Four-Ordered-H/index.html>.

Theorem 3.1: The graph CCC_5 is both 4-ordered hamiltonian and 4-ordered hamiltonian connected.

IV. DISCUSSION

In this paper, we provide a 3-regular vertex transitive graph that is 4-ordered hamiltonian and 4-ordered hamiltonian connected. With this example, 3-regular 4-ordered hamiltonian graphs are not rare as we expected. Thus, finding or classifying such graphs is more interesting. Once we have more knowledge of these graphs we may construct k -regular graphs that are $(k+1)$ -hamiltonian, $(k+1)$ -hamiltonian connected, or $(k+1)$ -hamiltonian laceable graphs with $k > 4$.

Can we add one more property, say edge transitive? In other words, we want to find a 3-regular, vertex transitive, edge transitive, 4-ordered hamiltonian, and 4-ordered hamiltonian connected graph. We cannot answer this question. However, we provide the following information.

The n -dimensional star graph, denoted by S_n , is a graph with the vertex set $V(S_n) = \{u_1 u_2 \dots u_n \mid u_i \in [n] \text{ and } u_i \neq u_j \text{ for } i \neq j\}$. The adjacency is defined as follows: $u_1 u_2 \dots u_i \dots u_n$ is adjacent to $v_1 v_2 \dots v_i \dots v_n$ through an edge of dimension i with $2 \leq i \leq n$ if $v_j = u_j$ for $j \notin \{1, i\}$, $v_1 = u_i$, and $v_i = u_1$. The n -dimensional star network S_n was proposed in [1] as “an attractive alternative to the n -cube topology for interconnecting processors in parallel computers. Using computer, we can prove that S_4 is a 3-regular, vertex transitive, edge transitive, and 4-ordered hamiltonian laceable. However, S_4 is not 4-ordered hamiltonian.

In mathematics, “one example” solves the “existence problem”. Yet, it may be a *unique solution*,

many but finite solutions, or *an infinite number of solutions*. We are wondering if there are an infinite number of 3-regular vertex transitive graphs that are 4-ordered hamiltonian and 4-ordered hamiltonian connected. More precisely, we are wondering if any CCC_n with n is odd and $n \geq 5$ is 4-ordered hamiltonian and 4-ordered hamiltonian connected. Furthermore, any CCC_n with n is even and $n \geq 6$ is 4-ordered hamiltonian and 4-ordered hamiltonian laceable.

Finally, we note that the Heawood graph, $CR(14, 1, 5)$, is the only known 3-regular bipartite graph that is 4-ordered hamiltonian but not 4-ordered hamiltonian laceable. The generalized Petersen graph, $P(20, 4)$, is the only known 3-regular graph that is 4-ordered hamiltonian connected but not 4-ordered hamiltonian. Moreover, the star graph, S_4 , is the only known 3-regular, vertex transitive, edge transitive, and 4-ordered hamiltonian laceable graph that is not 4-ordered hamiltonian. We are wondering if these graphs are “unique” in their own corresponding families.

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